

Feature Selection (Independent Variables) and Target Variable

Feature Selection

Before training the machine learning model, the dataset is divided into **independent variables (X)** and the **dependent variable (Y)**.

Independent Variables (X):

These are the input features used by the model to predict the Human Development Index (HDI). The selected columns are: Column Index 2 – Country Column Index 5 – Life Expectancy Column Index 6 – Expected Years of Schooling Column Index 7 – Mean Years of Schooling Column Index 67 – Gross National Income (GNI) per Capita All selected features are stored in **X** as a Pandas DataFrame.

Dependent Variable (Y):

The target variable is the **Human Development Index (HDI)**, located at **Column Index 4**. The HDI values are stored in **Y** as a Pandas DataFrame.

Code:

```
X = Development.iloc[:, [2, 5, 6, 7, 67]]  
X = pd.DataFrame(X)
```

```
y = Development.iloc[:, 4].values  
y = pd.DataFrame(y)
```

Summary:

The selected independent variables (Country, Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income per Capita) are used as inputs to predict the Human Development Index (HDI), which serves as the dependent variable.